

189 interactive simulations with real numerics — no install, no cloud, free forever.

The Wolfram Alpha of interactive simulation

Flagship Engines, Fields & GPU Compute ⁽¹⁹⁾

Visual Node Graph /studio/graph	Wire blocks into a live dataflow — the real editor. Symbolic calculus built in.
2D Fluid (CFD) /studio/fluid	Interactive incompressible Navier–Stokes via Stam's stable-fluids method.
Heat & Wave Fields /studio/fields	2D heat diffusion and the 1D wave equation, solved live.
AI Surrogate /studio/surrogate	Train an ML surrogate on our solver for instant predictions. The PhysicsX play.
GPU Compute (WebGPU) /studio/gpu	Hundreds of thousands of particles on your GPU via WGSL shaders.
WebGPU Fluid /studio/gpu-fluid	High-res GPU smoke advection you can stir.
GPU N-Body /studio/gpu-nbody	Thousands of bodies, full $O(n^2)$ gravity on the GPU.
GPU PDE Solver /studio/gpu-pde	Steady heat/Poisson via GPU Jacobi at high res.
GPU 3D Fluid /studio/gpu-fluid-3d	3D smoke, simulated + raymarched on the GPU.
Particle-Mesh N-Body /studio/gpu-nbody-pm	100k+ bodies via GPU particle-mesh gravity.
3D CFD /studio/cfd-3d	3D Navier–Stokes plume with z-slices.
3D N-Body /studio/3d	Gravitation in 3D with a drag-to-orbit camera.
3D FEA Space Frame /studio/fea-3d	3D structural analysis; orbit the deformed tower.

3D Heat Diffusion /studio/heat-3d	Volumetric heat with slices and orbit view.
FEA Truss /studio/fea	Finite-element structural analysis: forces & deflection.
Meshing + BCs /studio/mesh	Paint boundary conditions; solve steady heat.
Optimize + UQ /studio/optimize	Gradient descent and Monte-Carlo uncertainty.
RLC Circuit /studio/rlc	Step response and damping regimes.
Gradient Descent /studio/gradient-descent	Optimization on loss landscapes.

Physics, Mechanics & Thermodynamics ⁽²⁵⁾

Particle / N-Body /studio/particles	Gravity, orbits, and impulse collisions with a real symplectic integrator.
Electrostatics /studio/electromagnetics	Charges, potential heatmaps, and field lines.
Double Pendulum /studio/double-pendulum	The textbook chaotic system, integrated with RK4.
Projectile Motion /studio/projectile	Ballistics with air drag; tune angle and speed.
Ising Model /studio/ising	A live magnetic phase transition via Monte Carlo.
Strange Attractors /studio/attractors	Lorenz, Rössler, Aizawa & more in 3D.
Wave Interference /studio/wave-interference	Two-source ripple-tank fringes.
Random Walk /studio/random-walk	Brownian motion and diffusion.
Kepler Orbits /studio/kepler	Conic-section orbits under gravity.

Double-Slit /studio/double-slit	Quantum interference + diffraction.
Cloth / Spring-Mass /studio/cloth	Verlet cloth you can grab and swing.
Gravity Well /studio/gravity-well	Curved-spacetime rubber sheet + orbit.
Magnetic Pendulum /studio/magnetic-pendulum	Fractal basins of attraction.
Percolation /studio/percolation	A phase transition in connectivity.
DLA Growth /studio/dla	Branching fractals from random walks.
Ray Optics / Lenses /studio/ray-optics	Trace rays through lenses.
Gas in a Box /studio/gas	Kinetic theory → pressure ($PV=nRT$).
Elastic Collisions /studio/collisions	Momentum & energy conservation.
Buoyancy /studio/buoyancy	Archimedes — float or sink by density.
Rocket Equation /studio/rocket	Tsiolkovsky Δv budget calculator.
Blackbody Radiation /studio/blackbody	Planck curve + Wien's law.
Pendulum Wave /studio/pendulum-wave	Phase art from many periods.
Orbital Transfer /studio/orbital-transfer	Hohmann transfer Δv .
Standing Waves /studio/standing-waves	Harmonics, nodes & antinodes.

Doppler Effect</studio/doppler>

Moving-source wavefronts & sonic boom.

Waves, Optics & Acoustics (14)**Snell's Law**</studio/snells-law>

Refraction & total internal reflection.

Diffraction Grating</studio/diffraction-grating>

N-slit interference maxima.

Beats</studio/beats>

Two close tones interfere.

Curved Mirrors</studio/mirror>

Concave/convex ray tracing.

Prism Dispersion</studio/prism>

White light into a spectrum.

Fourier Transform</studio/fft>

Signal → frequency spectrum.

Harmonic Series</studio/harmonic-series>

Overtones & timbre.

Equal Temperament</studio/equal-temperament>

Tuning & cents.

Chladni Plates</studio/chladni>

Nodal sand patterns.

Helmholtz Resonator</studio/helmholtz-resonator>

Cavity resonance.

Room Modes</studio/room-modes>

Standing waves in a room.

Reverb Time (Sabine)</studio/reverb-time>

RT60 decay.

Sound Levels (dB)</studio/sound-levels>

Decibel addition & falloff.

Additive Synthesis</studio/additive-synthesis>

Tone from sine waves.

Mathematics (22)

Dynamical Systems /studio/dynamics	Lorenz, SIR epidemics, pendulums, predator–prey, and reaction–diffusion.
Symbolic Math /studio/cas	Parse, differentiate, simplify, solve, and plot — a real CAS in your browser.
Vector Fields /studio/vector-field	Plot any $F(x,y)$ as a live quiver diagram.
Notebook /studio/notebook	Cells of prose + symbolic math + compute.
Fractal Explorer /studio/fractals	Zoom into the Mandelbrot and Julia sets.
Fourier Series /studio/fourier	Build waves from sine harmonics.
Function Grapher /studio/grapher	Plot up to three functions at once.
3D Surface Plotter /studio/surface-3d	Orbit any $z = f(x, y)$ surface.
Matrix Calculator /studio/matrix	Multiply, invert, determinant, transpose.
Cellular Automata /studio/cellular-automata	Wolfram rules & Conway's Life.
Taylor Series /studio/taylor	Polynomial approximation, live.
Newton's Method /studio/newton	Root finding, tangent by tangent.
Distributions /studio/distributions	Normal, Poisson, binomial & more.
Complex Functions /studio/complex	Domain coloring of $f(z)$.
Lissajous Curves /studio/lissajous	Two sine waves, endless patterns.

Bifurcation Diagram /studio/bifurcation	Logistic map route to chaos.
Direction Field /studio/direction-field	Slope fields + solution curves.
Riemann Sums /studio/riemann	Integrals as stacked rectangles.
Eigenvectors /studio/eigenvectors	2×2 transforms & invariant directions.
Markov Chains /studio/markov	Converge to a stationary distribution.
Parametric Grapher /studio/parametric	Trace $x(t)$, $y(t)$ curves.
Euler vs RK4 /studio/numerical-methods	ODE integration accuracy.

Statistics & Data Science (8)

Hypothesis Test /studio/hypothesis-test	p-values & rejection regions.
Linear Regression /studio/linear-regression	Least squares & R^2 .
Central Limit Theorem /studio/central-limit	Means go normal.
Confidence Intervals /studio/confidence-interval	What 95% really means.
Bayesian Inference /studio/bayes-inference	Prior × likelihood → posterior.
Bootstrap /studio/bootstrap	Resampling confidence.
PCA /studio/pca	Axes of maximum variance.
A/B Test /studio/ab-test	Two-proportion significance.

Chemistry & Materials (8)

Molecular Dynamics /studio/molecular-dynamics	Lennard-Jones atoms; melt a lattice.
Titration Curve /studio/titration	pH vs base added; equivalence point.
Reaction Kinetics /studio/reaction-kinetics	Rate laws & Arrhenius decay.
Chemical Equilibrium /studio/equilibrium	Le Chatelier shifts.
pH Calculator /studio/ph	pH/pOH and the acid-base scale.
Radioactive Decay /studio/radioactive-decay	Half-life exponential decay.
Maxwell-Boltzmann /studio/maxwell-boltzmann	Molecular speed distribution.
Ideal Gas Law /studio/ideal-gas	$PV = nRT$ with a live piston.

Computer Science & AI (10)

Sorting Visualizer /studio/sorting	Watch algorithms sort in real time.
Boids Flocking /studio/boids	Emergent flocking from three rules.
A* Pathfinding /studio/pathfinding	Shortest path, draw walls.
Maze Generator /studio/maze	Carve & solve perfect mazes.
Neural Network /studio/neural-net	Train an MLP, watch it learn.
k-Means Clustering /studio/kmeans	Unsupervised clustering, live.

Convex Hull /studio/convex-hull	Wrap points in $O(n \log n)$.
L-System Fractals /studio/l-system	Grow plants from grammars.
Turing Machine /studio/turing-machine	The model of computation.
Image Convolution /studio/convolution	Kernels behind every CNN.

Complex Systems & Chaos ⁽⁹⁾

Traffic Flow /studio/traffic	Phantom jams from random braking.
Forest Fire Model /studio/forest-fire	Self-organized criticality.
Abelian Sandpile /studio/sandpile	Avalanches build a fractal.
Schelling Segregation /studio/schelling	Mild bias, sharp segregation.
Langton's Ant /studio/langtons-ant	Two rules, emergent highway.
Reaction-Diffusion /studio/reaction-diffusion	Gray-Scott Turing patterns.
Elementary CA /studio/wolfram-ca	Wolfram's 256 rules.
Genetic Algorithm /studio/genetic-algorithm	Evolve toward the peak.
Ant Colony Optimization /studio/ant-colony	Swarm-solved TSP.

Quantum & Modern Physics ⁽⁸⁾

Particle in a Box /studio/particle-box	Quantized energy levels.
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Quantum Tunneling /studio/quantum-tunneling	Through a forbidden barrier.
Bloch Sphere /studio/bloch-sphere	The state of a qubit.
Hydrogen Orbitals /studio/hydrogen-orbitals	Electron probability clouds.
Special Relativity /studio/special-relativity	Time dilation & contraction.
Photoelectric Effect /studio/photoelectric	Einstein's photon.
Quantum Harmonic Oscillator /studio/quantum-harmonic	Evenly-spaced energy ladder.
Stern-Gerlach /studio/stern-gerlach	Spin measurement.

Biology & Medicine ⁽¹⁰⁾

Epidemic Network /studio/epidemic-network	Agent-based SIR on a contact graph.
Spatial Predator–Prey /studio/predator-prey	Agent-based ecology with waves.
SIR Epidemic /studio/sir-model	Outbreak compartment model.
Hodgkin-Huxley Neuron /studio/neuron	Action potential spikes.
Lotka-Volterra /studio/lotka-volterra	Predator-prey cycles.
Hardy-Weinberg /studio/hardy-weinberg	Allele frequency & selection.
Enzyme Kinetics /studio/enzyme-kinetics	Michaelis-Menten curves.
Logistic Growth /studio/logistic-growth	Population carrying capacity.

Genetic Drift</studio/genetic-drift>

Wright-Fisher allele drift.

Sequence Alignment</studio/sequence-alignment>

Needleman-Wunsch DNA.

Astronomy & Space (8)**Exoplanet Transit**</studio/exoplanet-transit>

Light-curve transit method.

H-R Diagram</studio/hr-diagram>

The map of the stars.

Lagrange Points</studio/lagrange-points>

Restricted 3-body equilibria.

Roche Limit</studio/roche-limit>

When tides shred a moon.

Hubble's Law</studio/hubble-law>

The expanding universe.

Telescope Optics</studio/telescope>

Resolution & magnification.

Stellar Parallax</studio/parallax>

Distance from angle shift.

Escape Velocity</studio/escape-velocity>

Speed to leave a world.

Earth, Climate & Environment (8)**Energy Balance**</studio/energy-balance>

Greenhouse equilibrium temp.

Daisyworld</studio/daisyworld>

Planetary self-regulation.

Milankovitch Cycles</studio/milankovitch>

Orbital forcing of ice ages.

Tsunami Propagation</studio/tsunami>

Shallow-water wave shoaling.

Carbon Cycle /studio/carbon-cycle	Reservoir box model.
Earthquake Shaking /studio/seismic	Magnitude, energy, intensity.
Groundwater Drawdown /studio/groundwater	Well cone of depression.
Lapse Rate /studio/lapse-rate	Atmosphere & cloud base.

Finance & Quantitative ⁽⁸⁾

Black-Scholes /studio/black-scholes	Option pricing + Greeks.
Option Payoff /studio/option-payoff	Strategy P/L diagrams.
Monte Carlo (GBM) /studio/monte-carlo	Price-path distributions.
Efficient Frontier /studio/efficient-frontier	Portfolio risk vs return.
Value at Risk /studio/value-at-risk	VaR & expected shortfall.
Bond Pricing /studio/bond-pricing	Price, yield & duration.
Compound Interest /studio/compound-interest	Investment growth over time.
Loan Amortization /studio/amortization	Mortgage payment breakdown.

First Responders (Fire | Police | EMS | Hazmat) ⁽⁸⁾

Wildfire Spread /studio/fire-spread	Wind + slope fire behavior.
Hazmat Plume /studio/hazmat-plume	Gaussian dispersion + PAD.

Building Evacuation /studio/evacuation	Egress flow & clear time.
START Triage /studio/triage	Mass-casualty triage tool.
Fire Hose Hydraulics /studio/hose-flow	Friction loss & pump pressure.
Skid-to-Stop Speed /studio/skid-to-stop	Accident reconstruction.
Radio Range /studio/radio-range	VHF/UHF link budget.
Blast Standoff /studio/blast-standoff	Overpressure & safe distance.

Structural & Civil Engineering (8)

Beam Deflection /studio/beam-deflection	Sag & bending moment.
Column Buckling /studio/column-buckling	Euler critical load.
Mohr's Circle /studio/mohrs-circle	2D stress transformation.
Shear & Moment /studio/shear-moment	SFD/BMD diagrams.
Retaining Wall /studio/retaining-wall	Rankine earth pressure.
Seismic Base Shear /studio/base-shear	Equivalent lateral force.
Soil Bearing /studio/soil-bearing	Terzaghi capacity.
Concrete Beam /studio/concrete-beam	RC flexural capacity.

Electrical & Signal Processing ⁽⁸⁾

Bode Plot /studio/bode-plot	Frequency response.
Filter Designer /studio/filter-designer	RC/RLC filter response.
PID Controller /studio/pid-control	Step-response tuning.
Transmission Line /studio/transmission-line	VSWR & reflections.
Op-Amp Circuits /studio/op-amp	Amplifier configurations.
Three-Phase Power /studio/three-phase	Grid power & phasors.
Transistor Bias /studio/transistor-bias	BJT Q-point load line.
Sampling & Aliasing /studio/aliasing	The Nyquist theorem.

Robotics, Control & Kinematics ⁽⁸⁾

Forward Kinematics /studio/forward-kinematics	Joint angles → end effector.
Inverse Kinematics /studio/inverse-kinematics	Target → joint angles.
Differential Drive /studio/differential-drive	Wheel speeds → path.
Cart-Pole Balance /studio/cart-pole	Inverted pendulum control.
Quadcopter Control /studio/quadcopter	Nested PID flight.
RRT Path Planning /studio/rrt	Sampling-based planner.

DC Motor</studio/dc-motor>

Speed-torque & step.

Kalman Filter</studio/kalman-filter>

Optimal state estimation.